I keep getting a mismatch error but i don't know why?

No worries! We can figure it out together. 😊

A **mismatch error** in Python often means that two things don't match in the way Python or your assignment expects. It could involve data types, values, variable names, or even expected output.

Let's narrow it down.

First, can you share two things?

1. The exact error message you're getting (copy and paste it if possible).
2. The part of your Python code where the error happens.

I'll help you trace through it rather than just rewrite your code.

One question to start: **Does the error say `TypeError`, `ValueError`, or something like `expected output does not match`?**

We'll take it one step at a time!

Type Mismatch

Your code tried to use a value in a way that does not work for its type (for example, mixing text and numbers).

request.route_execution_failed: Error while processing request for URL 'save_assignment': can only concatenate str (not "int") to str

What To Try Next

- Check that each function call has the right number of arguments.
- Verify that values have the type your code expects (for example, text vs number).
- Print intermediate values to see what type they are before the failing line.

Navigation Options:

- Return to Index Page
- Reset State and Return to Index
- Reload Entire Application
- Submit a Bug Report

Technical Details

Error ID	request.route_execution_failed
Message	Error while processing request for URL 'save_assignment': can only concatenate str (not "int") to str
Route Call	save_assignment(State(assignments=[], course_name='cisc', current_grade=0.0, target_grade=90.0, needed_grade=0.0, message='Course name saved.'), 'exam 1', 70.0)
Status	error
Severity	error
Category	request
Recoverable	yes
Request	Show details

id	4
action	link
url	save_assignment
kwargs	

assignment_name	exam 1
grade	70
--submit-button	"Save Assignment#drafter-component-37418792"

event	{}
dom_id	drafter-component-37418792
button_pressed	drafter-component-37418792
raw_payload	

-

name	assignment_name
value	exam 1

source `form_field`

source_detail

1.

name `grade`

value `70`

source `form_field`

source_detail

1.

name `--submit-button`

value `"Save Assignment#drafter-component-37418792"`

source `form_field`

source_detail

Traceback

Read from the bottom up: the last line names the error, and the nearest “your code” entry above it shows where to look.

Traceback (most recent call last):

File `/lib/python3.14/site-packages/drafter/client_server/client_server.py` , line `523` , in `execute_route`

return route_func(*args, **kwargs), representation ~~~~~~~~~~^^^^^^^^^^^^^^^^^

File `unsaved_thonny_script.py` , line `113` , in `save_assignment` **your code**

return view_assignments(state)

File `unsaved_thonny_script.py` , line `126` , in `view_assignments` **your code**

assignment.name + ":" + int(assignment.grade) + "%"
~~~~~~~~~~~~~~~~~~~~~~^~~~~~~~~~~~~~~~~~~~~~~
~~~~~~~~~~~~~~~~~~~~~~

TypeError: can only concatenate str (not "int") to str

Route:save_assignment❌ 1📌 Persisted (0)🖨️ Console

🖨️ **Console**ClearHide

Pillow already loaded from default channel Loading Jinja2, MarkupSafe, contourpy, cycler, fonttools, kiwisolver, matplotlib, numpy, packaging, pyparsing, python-dateutil, pytz, six Loaded Jinja2, MarkupSafe, contourpy, cycler, fonttools, kiwisolver, matplotlib, numpy, packaging, pyparsing, python-dateutil, pytz, six TEST PASSED - [line 169] assert_equal(average_grades(test_assignments), 80.0) TEST PASSED - [line 170] assert_equal(average_grades([]), 0.0)

>>>

Debug Panel

View

Current1HistoryOverviewTestsEnvironment

Current Page

Route: `save_assignment`

Page Content

Errors & Warnings

❌Your code tried to use a value in a way that does not work for its type (for example, mixing text and numbers).Details

🔍 Compare with previous state

Current State

State

assignments:

[]

Empty list

course_name:

'cisc'

str

current_grade:

0.0

float

target_grade:

90.0

float

needed_grade:

0.0

float

message:

'Course name saved.'

str

Great, this traceback gives us a specific place to investigate. The important part is:

```
None
assignment.name + ":" + int(assignment.grade) + "%"
```

The error says:

`can only concatenate str (not "int") to str`

What Python is telling you

Python is evaluating the `+` operators from left to right here.

- `assignment.name` is likely a string.
- `":"` is a string.
- `int(assignment.grade)` produces an integer.

- `"%"` is a string.

The issue occurs when Python tries to combine a string with an integer using `+`. The `+` operator can concatenate strings, but it cannot directly concatenate a string and an integer.

Your textbook explains that **types determine which operations are allowed**. For example, strings can be joined with other strings, but you cannot subtract strings or combine a string and a number through string concatenation.

Let's reason it through

Before changing anything, consider these questions:

1. What type does `int(assignment.grade)` produce?
2. What type does Python expect if you're using `+` to join that value to `assignment.name`?
3. Do you want the grade to be used as a number for a calculation, or displayed as text on the page?

One more question: What is the purpose of `view_assignments` in your program—are you trying to display each assignment's name followed by its grade and a percent sign?

Your answers will help us decide what needs to change without accidentally affecting the grade calculations elsewhere.

```
from dataclasses import dataclass
from drafter import *
from bakery import assert_equal


@dataclass
class Assignment:
    """Store the name and grade for one assignment."""
    name: str
    grade: float


@dataclass
class State:
    """Store information that the website needs between pages."""
    assignments: list[Assignment]
    course_name: str
```

```
    current_grade: float
    target_grade: float
    needed_grade: float
    message: str


def average_grades(assignments: list[Assignment]) -> float:
    """Calculate the average of all assignment grades."""
    if len(assignments) == 0:
        return 0.0

    total = 0.0
    for assignment in assignments:
        total = total + assignment.grade

    return total / len(assignments)


def required_next_grade(assignments: list[Assignment],
                        target: float) -> float:
    """Calculate the grade needed on the next assignment."""
    total = 0.0

    for assignment in assignments:
        total = total + assignment.grade

    number_of_assignments = len(assignments) + 1
    needed = target * number_of_assignments - total

    return needed


def valid_grade(grade: float) -> bool:
    """Check whether a grade is between 0 and 100."""
    return grade >= 0 and grade <= 100


@route
def index(state: State) -> Page:
    """Display the home page and main navigation."""
    return Page(state, [
        "COLLEGE GRADE CALCULATOR",
        "Welcome! Use this website to keep track of your grades.",
        "Course name: " + state.course_name,
```

```
        TextBox("course_name", state.course_name),
        Button("Save Course Name", "save_course"),
        " ",
        Button("Add Assignment", "add_assignment"),
        Button("View Assignments", "view_assignments"),
        Button("Calculate Current Grade", "calculate_grade"),
        state.message
    ])


@route
def save_course(state: State, course_name: str) -> Page:
    """Save the course name entered by the student."""
    if course_name.strip() == "":
        state.message = "Please enter a course name."
    else:
        state.course_name = course_name
        state.message = "Course name saved."

    return index(state)


@route
def add_assignment(state: State) -> Page:
    """Display the form for entering a new assignment."""
    return Page(state, [
        "ADD AN ASSIGNMENT",
        "Enter the assignment name:",
        TextBox("assignment_name"),
        "Enter the assignment grade (0-100):",
        TextBox("grade", 0.0, "number"),
        Button("Save Assignment", "save_assignment"),
        Button("Home", "index")
    ])


@route
def save_assignment(state: State, assignment_name: str,
                    grade: float) -> Page:
    """Check and save a new assignment."""
    if assignment_name.strip() == "":
        state.message = "Please enter an assignment name."
    elif not valid_grade(grade):
        state.message = "Grade must be between 0 and 100."
```

```
    else:
        new_assignment = Assignment(assignment_name, grade)
        state.assignments.append(new_assignment)
        state.current_grade = average_grades(state.assignments)
        state.message = "Assignment successfully added."

    return view_assignments(state)


@route
def view_assignments(state: State) -> Page:
    """Display all assignments currently saved."""
    content = ["YOUR ASSIGNMENTS"]

    if len(state.assignments) == 0:
        content.append("You have not entered any assignments yet.")
    else:
        for assignment in state.assignments:
            content.append(
                assignment.name + ":" + int(assignment.grade) + "%"
            )

    content.append(" ")
    content.append("Current average: " +
                   int(round(average_grades(state.assignments), 2)) + "%")
    content.append(state.message)
    content.append(Button("Add Another Assignment", "add_assignment"))
    content.append(Button("Calculate Current Grade", "calculate_grade"))
    content.append(Button("Home", "index"))

    return Page(state, content)


@route
def calculate_grade(state: State) -> Page:
    """Calculate and display the student's current grade."""
    if len(state.assignments) == 0:
        state.message = "Add assignments before calculating your grade."
        state.current_grade = 0.0
    else:
        state.current_grade = average_grades(state.assignments)
        state.message = "Your current grade has been calculated."

    return Page(state, [
```

```
        "YOUR CURRENT GRADE",
        "Course: " + state.course_name,
        "Current average: " + str(round(state.current_grade, 2)) + "%",
        state.message,
        Button("View Assignments", "view_assignments"),
        Button("Home", "index")
    ])



# Tests for the calculation and validation functions
test_assignments = [
    Assignment("Homework 1", 80.0),
    Assignment("Quiz 1", 90.0),
    Assignment("Exam 1", 70.0)
]

assert_equal(average_grades(test_assignments), 80.0)
assert_equal(average_grades([]), 0.0)


start_server(State([], "", 0.0, 90.0, 0.0, ""))
```

Thanks for sharing your code! I can see exactly where the error is coming from. Let's look at the types involved rather than rewriting the function for you.

1. The first mismatch

Inside `view_assignments`, you have:

```
None
assignment.name + ":" + int(assignment.grade) + "%"
```

Think about the types of each part:

Expression	Type

`assignment.name`	`str`
`":"`	`str`
`int(assignment.grade)`	`int`
`"%"`	`str`

Python's `+` operator concatenates strings, but it cannot concatenate a string and an integer. That's why you see the `TypeError`.

2. There's another similar mismatch

A little farther down, you have:

```
None
"Current average: " + int(round(average_grades(state.assignments), 2)) + "%"
```

Let's trace the types:

- `"Current average: "` is a string.
- `average_grades(...)` returns a float.
- `round(..., 2)` returns a number.
- `int(...)` converts that number to an integer.

So this expression has the same kind of mismatch.

3. Your turn: reason about the conversion

You already use `str()` successfully in `calculate_grade`:

```
None
"Current average: " + str(round(state.current_grade, 2)) + "%"
```

Compare that expression with the two expressions in `view_assignments`.

A couple of questions:

1. What does `str()` do to a numeric value?
2. Why might `str()` be more appropriate than `int()` when you want to display a grade such as `87.5%`?
3. Why would converting a grade to an integer potentially lose information?

The textbook's section on types explains that a value's type determines which operations Python allows. Here, the important distinction is between a number used for calculations and text used for display.

Try making your own change to the two expressions in `view_assignments`. Then test the page by saving an assignment with a grade of `70.5`. What do you observe?

Annotations

1 error, 3 warnings, and 2 notices

build	**Process completed with exit code 1.**
build	Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: astral-sh/setup-uv@v6. For more information see: https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/
build	No file matched to [/home/runner/work/college-course-grader/college-course-grader/**/*requirements*.txt,/home/runner/work/college-course-grader/college-course-grader/**/*requirements*.in,/home/runner/work/college-course-grader/college-course-grader/**/*constraints*.txt,/home/runner/work/college-course-grader/college-course-grader/**/*constraints*.in,/home/runner/work/college-course-grader/college-course-grader/**/pyproject.toml,/home/runner/work/college-course-grader/college-course-grader/**/uv.lock,/home/runner/work/college-course-grader/college-course-grader/**/*.py.lock]. The cache will never get invalidated. Make sure you have checked out the target repository and configured the cache-dependency-glob input correctly.
deploy	Node.js 20 is deprecated. The following actions target Node.js 20 but are being forced to run on Node.js 24: actions/deploy-pages@v4. For more information see: https://github.blog/changelog/2025-09-19-deprecation-of-node-20-on-github-actions-runners/

build	"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/actions/runner-images/issues/14748"
deploy	"The ubuntu-latest label will migrate to Ubuntu 26 beginning October 19, 2026. For more information, see https://github.com/actions/runner-images/issues/14748"

Artifacts

Good news: these annotations show several warnings, but **they don't all indicate problems with your Python code**. Let's separate them.

1. The important message: build failed

Build failed

The key line is:

```
Process completed with exit code 1
```

This tells us that the build process failed, but it does not tell us why. We need the actual error earlier in the build log.

2. The other annotations

Annotation	What it means
Node.js 20 is deprecated	A GitHub Actions maintenance warning.
No file matched requirements	The dependency cache couldn't find a matching requirements or project configuration file.
Ubuntu 26 migration	A notice about a future runner change.

These messages are not, by themselves, proof of a Python error.

3. Let's investigate the actual failure

Your earlier Python error happened in `view_assignments`, but these GitHub Actions annotations don't show whether that issue is still present.

Open the failed **build** job and look at the log around the first red `Error` message, especially steps after the dependency setup.

Can you paste the first actual error message from the build log?

One useful distinction to keep in mind: a warning is not necessarily a failure. We want to find the message that explains why the build exited with code 1, rather than changing things based on unrelated warnings.